

PARTH GOYAL

Quantitative Researcher · Economics Writer

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Education

University of St Andrews

Sep 2020 – Jun 2024

BSc (Honours) Financial Economics and Mathematics, Upper Second Class (2:1)

St Andrews, UK

- **Dissertation (Mathematics):** *Modeling the Divide* — built a 5,000-agent heterogeneous ABM (partisan/elite types, ANES-initialised opinions, Gini-scaled tolerance) showing exposure and initial-distribution asymmetry drive bimodal polarization [\[live app\]](#)

CFA Institute

CFA Level I — Passed (Top 5% globally)

Experience

Orbis Investments

Sep 2024 – Present

Quantitative Researcher

London, UK

- Designed and own systematic stock-selection signals for global financials, using fundamental line-items to predict forward returns; backtests at ~10% annual outperformance and used in production for market-neutral portfolio construction (~\$1bn AUM)
- Built a novel time-series factor model for a concentrated US equity portfolio explaining ~80% of return variance (adj. R^2); used by the portfolio manager to diagnose bets and size hedges
- Use LLM APIs (Claude, GPT) to extract structured signal at scale from earnings-call transcripts and analyst reviews — a text-derived signal layer the team couldn't easily reach manually

Orbis Investments

Jun 2023 – Sep 2023

Quantitative Analyst — Summer Internship

London, UK

- Developed 4 advanced security-beta estimators (return winsorization, time-decay, Bayesian) for beta hedging; evaluated 24 beta types over 40 years of US equity data, achieving ~1% lower volatility than the firm's prior standard (R)
- Engineered 50+ features from PMs' historical trades to capture behavioural biases; built an interpretable XGBoost model to flag likely losing trades (SHAP, walk-forward validation)

Risidio (Web3 startup)

May 2022 – Jul 2022

Finance & Investments Intern

London, UK

- Built NFT pricing models (comparables, probability distributions); analysed on-chain data and back-tested strategies to establish the company's data-analytics function
- Contributed to a decentralized film platform developed with Hollywood directors Lucas Foster and Eric Blakeney

Research

University of St Andrews

Sep 2021 – May 2022

Undergraduate Research Assistant — Vertically Integrated Project

St Andrews, UK

- Built a Python logistic model of the St Andrews housing market predicting equilibrium rent and demand from income, degree type, and housing features; calibrated parameters with `scipy.optimize`
- Developed a labour search-and-match model to explain wage rigidity from empirical evidence

Leadership & Founding

St Andrews AI Safety (STAISH)

2024

Co-founder & Program Director

St Andrews, UK

- Co-founded the University of St Andrews' first AI safety fellowship — a 6-week program covering ML fundamentals, AGI risk, alignment, and governance
- Curated 30+ readings (Ngo, Hubinger, Yudkowsky, DeepMind, OpenAI, Dafoe) and facilitated weekly discussions; outcomes include a participant taking an OpenAI internship

St Andrews Investment Society

Oct 2022 – Jun 2024

Quant Strategist

St Andrews, UK

- Built a CAPM / Sharpe-ratio toolkit with interactive Plotly dashboards; developed an APT model (rates, bond spreads, gold) for Sharpe-maximising portfolio construction

St Andrews Effective Altruism Society

Sep 2022 – Jun 2024

Treasurer & Tutor

St Andrews, UK

- Secured a £12,000 grant and ran cost-benefit fund allocation; led discussions and weekly workshops (forecasting, argument framing, charity evaluation) — 50% growth in enrolment

Spring Programs & Competitions

- **Spring insight programs:** Citadel & Citadel Securities, Jane Street, and J.P. Morgan (2021–2022)
- **Citadel Europe Datathon (2022):** EDA, statistical testing and ML on supply-chain data — improved shipping-time prediction by 30%
- **Quant trading competitions:** Citadel X Amplify Challenge and Traders at Berkeley — market-making, Bayesian updating, and risk management

Extracurricular & Community

- **Athletics:** national-level basketball (represented India and Russia, U18); state-level football and athletics (India); Men's 1st team basketball, University of St Andrews
- **Community:** built community toilets in rural Rajasthan; coached underprivileged students for a charity mini-marathon (3 finished top-10); 2025 half-marathon fundraiser for Concern Worldwide

Skills

Technical: Python [Pandas, NumPy, SciPy, scikit-learn, XGBoost, SHAP, cvxpy, LLM APIs], **R** [tidyverse, ggplot2, fixest, time-series], **SQL** (Snowflake), **LaTeX**
Languages: Russian, English, Hindi